

NIRMA UNIVERSITY
Institute of Technology
MCA Computer Science and Engineering
Practical List
Semester-I

6CS105CC26	Computer Networks	[3-0-2-4]
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Sr. No.	Practical Title	Hours	CLO
1	i) Identification of various network-related components. (using Internet) - Connections, RJ-45, BNC connector - Cables, Co-axial, twisted pair, UTP, Cross & Straight Cable - NIC (network interface card), Fiber Optics cable - Switch, Hub, Router ii) Practice network-related commands like Ping, Ipconfig, Netstat, tracert etc.	04	1, 2
2	Write a sender and receiver side bit stuffing program in which a frame is to be simulated with header, payload and trailer. In that, demonstrate the bit stuffing concept. Use files to read payload and write the payload back to the file. Use C/C++/Java [Starting and ending flags/indicators are 01111110].	04	1, 2
3	Implementation of a local area network using Cisco Packet Tracer (include switch and Hub)	02	2, 4
4	Perform Switch Configuration: configuration of ports, VLAN setup, IP- address	02	2
5	Implementation of Network Server Configurations (DHCP, FTP) using Cisco Packet Tracer.	02	1, 2
6	i) Implementation of Network Server Configurations (DNS, HTTP) using Cisco Packet Tracer. ii) Implementation of Mail Server using Cisco Packet Tracer.	04	1, 2
7	Demonstrating the concept of subnetting in CISCO Packet Tracer.	04	3
8	Simulate the configuration of the router using Cisco Packet Tracer: Static routing and RIP routing configuration	04	3,4
9	Analyze the network traffic and packet details using Wireshark.	02	4
10	Implement a server that provides the current time using client-server socket programming in Java.	02	2